

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : BS-EE301/BS-EEE301 Biology for Engineers

UPID : 003524

Time Allotted : 3 Hours

Full Marks :70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) What refers to as "Staff of life"?
- (II) Carnauba is an example of _____ wax
- (III) Where can scientists obtain stem cells?
- (IV) Cancer is caused by.....
- (V) Which part of the compound microscope helps in gathering and focusing light rays on the specimen to be viewed?
- (VI) Name of a transport protein is.....
- (VII) The ability of the immune system to recognize self-antigens versus nonself antigen is an example of.....
- (VIII)has a spiral metabolic pathway
- (IX) The longest-lasting immunity to an infectious agent is-----
- (X) Nuclear DNA replicates in the _____ phase.
- (XI) Name the inhibition where end products of biosynthesis pathway inhibit the activity of the first enzyme?
- (XII) Arrange the following sequences of tumor development in the correct order?
 - a) Metastasis
 - b) Progression
 - c) Promotion
 - d) Initiation

Group-B (Short Answer Type Question)*Answer any three of the following*

[5 x 3 = 15]

- 2. Discuss five Kingdoms of Classification. [5]
- 3. Define amino acid. What is the biological significance of it? [5]
- 4. What are the stages of cancer, and what do they mean? [5]
- 5. Write a short note on molecular basis of information transfer. [5]
- 6. Write a short note on Electrophysiology [5]

Group-C (Long Answer Type Question)*Answer any three of the following*

[15 x 3 = 45]

- 7. (a) Why Biology is as important as a scientific discipline like Mathematics, Physics and Chemistry? [5]
(b) Mention the most exciting aspect of biology as an independent scientific discipline. [10]
- 8. (a) Discuss about three major kingdoms of life. [3]
(b) Define spontaneity [2]
(c) Discuss ATP used as an energy currency. [10]
- 9. (a) What are enzymes? [2]
(b) Discuss about the factors affecting enzymatic activity. [8]
(c) What is Michelis-Menten constant? [2]
(d) Name three Cu containing enzymes. [3]
- 10. (a) What is Immunology? Briefly discuss the basic aspect of immunology. [5]
(b) Discuss about the basic aspect of pathogen. [5]
(c) Discuss about antigen-antibody reaction. [5]
- 11. (a) Define nucleic acid. [2]

(b) Briefly discuss about the monomers of nucleic acids.

[3]

(c) Write a short note on Nitrogenous base.

[10]